

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



Participant Details

- Participant Name: Devbrat Nayak
- Problem Statement: Create a data intelligence tool people would actually use
- Project: Bharat Ledger

Brief about the idea

Bharat Ledger - Turning 60-page budget PDFs into a 5-second verdict

- Public spending data is buried in Union Budget, state budget and CAG PDFs.
- Citizens search a project and instantly see two auditable scores: Justification and Transparency.
- A multi-agent AI pipeline debates evidence first, then shows the rationale behind the verdict.
- Built for citizens, students, journalists and RTI researchers who need plain-language evidence, not raw tables.

Approach with Google Cloud + NVIDIA

- Vertex AI + Gemini 2.5 Flash agents extract budget/CAG PDF claims: sanctioned, disbursed, utilized and outcomes.
- BigQuery stores structured spend data, cached verdicts and debate transcripts.
- ADK on Vertex AI orchestrates extraction, cross-check, Red Team / Blue Team debate and Council verdict.
- NVIDIA cudf.pandas accelerates state x sector x year ranking recomputation.

Workflow: PDF -> Extraction -> Cross-source check -> Debate -> Council verdict -> Dashboard scores.

Impact

- Citizens and journalists can judge spend reasonableness without engineering/economics expertise.
- Converts unreadable PDFs into comparable, plain-language scores in seconds.
- Shows source conflicts and rationale before asking users to trust a number.

Opportunities

- How different is it from any of the other existing ideas?
- USP of the proposed solution

Not another budget tracker - an auditable reasoning engine

- Existing transparency sites expose raw numbers; Bharat Ledger adds judgment support and visible reasoning.
- Separates two signals: is the spend reasonable, and do official sources agree?
- Frames the product as decision support, not a corruption tracker - less backlash, more actionable evidence.

USP

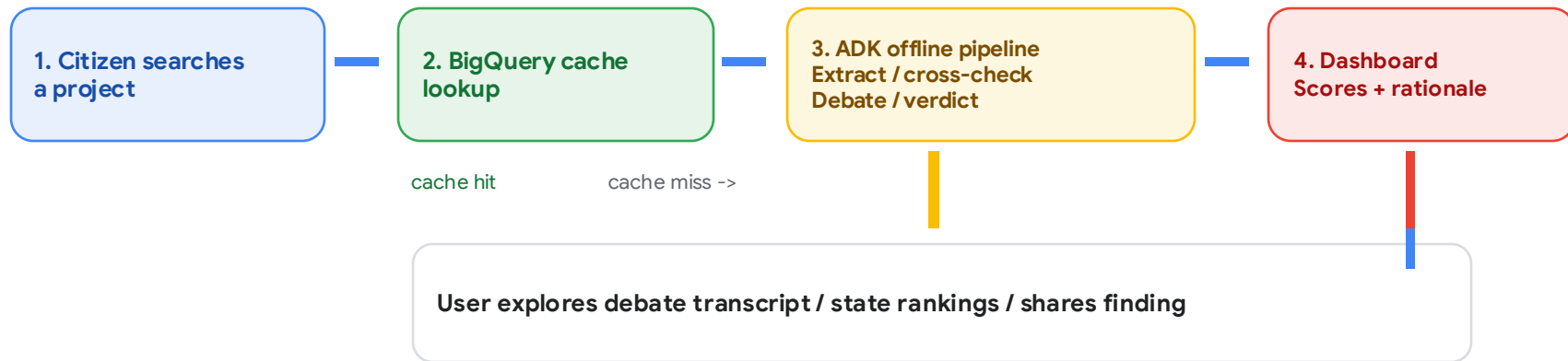
- Judicial-style ADK debate: Red Team, Blue Team and Council.
- Every score has a replayable rationale, not a black-box number.
- GPU-accelerated aggregation enables interactive state-vs-state comparisons.

List of features offered by the solution

What Bharat Ledger does

- Instant Justification Score - 0-100 reasonableness score per project with written rationale.
- Transparency / Confusion Score - flags source disagreement on basic facts.
- Multi-agent debate replay - watch Red Team vs. Blue Team, then Council verdict.
- State-vs-state rankings - spend-per-outcome comparisons across states and years.
- Roadmap: Photo-to-cost-check - upload an infrastructure photo for benchmark cost check.
- Roadmap: Citizen attention heatmap - search interest as a signal for stakeholders.

Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution

Product Walkthrough

[1] Search / home screen

Search bar + trending projects + state filter

[2] Project detail card

Justification Score gauge + Transparency badge + one-line rationale

[3] Debate replay view

Red Team argument | Blue Team argument

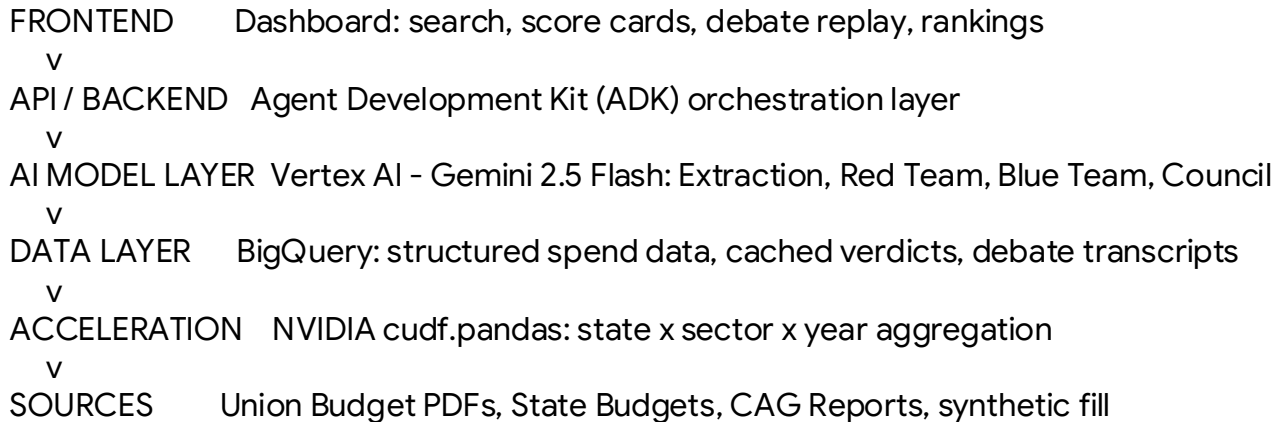
Council verdict banner below

[4] State ranking leaderboard

Sortable cost/outcome table with color-coded scores

Architecture diagram of the proposed solution:

System Architecture



Technologies / Google / Nvidia Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

Technology | Role | Why

- Vertex AI (Gemini 2.5 Flash) | Extraction, debate and verdict agents | Low-cost, scalable calls that draw on GCP credits.
- Agent Development Kit (ADK) | Multi-agent orchestration | Natural fit for debate-to-council reasoning.
- BigQuery | Structured data + cached verdicts | Avoids re-running agents for each dashboard request.
- NVIDIA cudf.pandas | Aggregation acceleration | Enables live state/sector/year ranking instead of static charts.

Scalability: expensive debate runs once offline per project; live users hit cached BigQuery verdicts.

Snapshots of the prototype

Required prototype snapshots

1. Search / home screen - search bar, trending projects, state filter
2. Project score card - Justification + Transparency scores with rationale
3. Debate replay view - Red vs. Blue arguments and Council verdict
4. State ranking leaderboard - sortable cost/outcome table with scores

Fallback status: use this list if final working screenshots are not ready before submission.

Google Cloud **H2S**

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Thank you

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